

UNIVERSITY OF CAPE TOWN
DEPARTMENT OF STATISTICAL SCIENCES - STA2020S: APPLIED STATISTICS
CLASS TEST 2 – 15 October 2025

Time: 90
MINUTES

Max Marks: 50

Internal Ms. A. Cloete
examiners: Mr Neil Watson

Instructions:

Answer ALL questions.

Marks are awarded for intermediate calculations.

Answer all MCQ questions *IN PENCIL* on the *MCQ sheet* provided.

Answer each written question *IN PEN* on a *NEW PAGE* in your answer book.

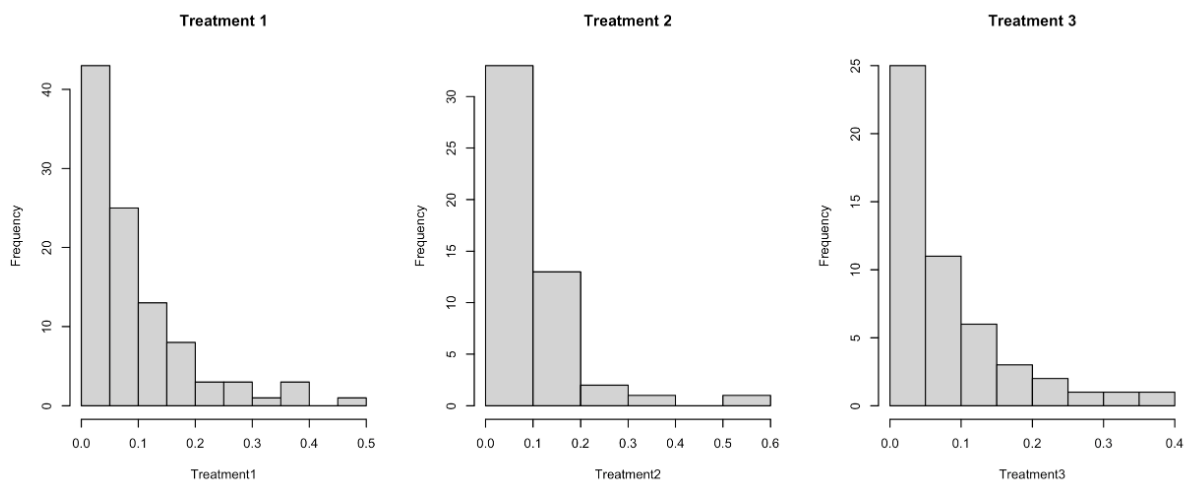
ROUND ALL ANSWERS TO 2 DECIMAL PLACES UNLESS OTHERWISE SPECIFIED.

DO NOT ROUND INTERMEDIARY CALCULATIONS.

SECTION A: MCQ QUESTIONS

[10]

1. An analyst was tasked with verifying whether data generated from a Completely Randomised Design with three treatments satisfies the assumptions of the corresponding ANOVA model. The following plots of the response variable by treatment were produced in R:



Choose the correct statement regarding the plots shown:

- A. The assumption being checked is about the normality of errors. The plots show that the responses are not normally distributed, so the errors are unlikely to be normal either. However, this should be rechecked after model fitting as well.
- B. The plots clearly show that the sample standard deviations are different which necessarily means that the assumption of equal population variance is violated.

- C. The treatments have not been randomised since the distributions differ across treatments.
 - D. There is a clear trend in the distribution of response per treatment (the frequency decreases with larger response values) this indicates that the assumption of independent errors has been violated. However, this should be checked after model fitting as well.
 - E. It is not sensible to check the assumptions of normality and independence using the response values as the assumptions are about the model errors.
2. A researcher conducted an experiment over a single day where participants were assigned to one of three treatments (A, B or C). Throughout the day, the researcher measured the response of participants who received:
- Treatment A in the morning,
 - Treatment B over lunch, and
 - Treatment C in the late afternoon.

Which of the following best describes the characteristics of the design used here?

- A. There is no replication within each treatment.
 - B. The researcher failed to randomise treatment order, so treatment is confounded with time of day.
 - C. The experiment should have used blocking by participant instead of randomisation.
 - D. The researcher should have used a paired-samples design to control for individual differences.
3. The experiment includes two factors, the specific treatment and time of day. When is it generally more appropriate to use a non-parametric test instead of a parametric test?
- A. When the sample sizes are very large.
 - B. When the population variances are equal.
 - C. When the data are ordinal and not normally distributed.
 - D. When there are no outliers.
 - E. When the data are measured on a ratio-scale.
4. COVID-19 had a huge impact on tourism in South Africa (RSA). Since 2022, tourism (measured by total monthly visitors to RSA) has grown steadily. Suppose you are interested in analysing the monthly visitors to your favourite beach in Cape Town. If you managed to obtain a time series that has this data since 2022, what time series components would you expect to see in a plot of the data? (2)
- A. Increasing trend, cyclical, random variation.
 - B. Trend, seasonality, cyclical and random variation.
 - C. Increasing trend, seasonal variation and random variation.
 - D. Increasing trend and random variation.
 - E. Seasonal variation, cyclical and random variation.
5. Which of the following are TRUE with regards to autocorrelation at lag 'g' in a time series dataset? (2)
- (i) It is a measure of the correlation between observations g time periods apart.
 - (ii) It can be removed from a time series using the Ljung-Box test.
 - (iii) It represents useful information that can be used by a model.

(iv) It is a good sign when we see autocorrelation in the residuals of a fitted model.

- A. (i) and (iv)
- B. (i)
- C. (ii) and (iii)
- D. (i), (iii) and (iv)
- E. (i) and (iii)

SECTION B: ANOVA QUESTIONS

[14]

QUESTION 1

[6]

Researchers are investigating whether light exposure on different parts of the body can reset the human circadian clock. Although jet lag usually fades as the brain's internal clock adjusts to new light–dark cycles through light entering the eyes, earlier studies suggested that light applied to the back of the knees might have a similar effect. To test this, 24 healthy volunteers were randomly assigned to one of three treatments:

1. *Eyes only* – exposed to light directed at the eyes for 3 hours,
2. *Knees only* – exposed to light directed at the back of the knees for 3 hours, or
3. *Control* – no light exposure during the same period.

The response variable was the *change in melatonin onset time (in hours)*, where negative values indicate a *delay* in melatonin production and positive values an *advance*. The researchers aimed to determine whether the mean melatonin shift differed among the three treatments. Below is some R output on the collected data.

1.1 Explain (in words) what a treatment effect is and calculate the estimated effect for the

```
> summary(jetlag_data)
  Treatment MelatoninShift
Control:8   Min.      :-0.9000
Eyes   :8   1st Qu.  :-0.5725
Knees   :8   Median  :-0.3150
          Mean     :-0.1646
          3rd Qu.  : 0.1400
          Max.     : 1.2400
> tapply(jetlag_data$MelatoninShift, jetlag_data$Treatment, mean)
Control      Eyes      Knees
-0.21625 -0.26125 -0.01625
> model <- aov(MelatoninShift ~ Treatment, data = jetlag_data)
> summary(model)
          Df Sum Sq Mean Sq F value Pr(>F)
Treatment  2  0.272   0.1361   0.398  0.677
Residuals 21  7.184   0.3421
```

Eyes only treatment. (2)

The treatment effects represents the average change in response for that treatment relative to the overall mean. / How much each treatment mean differs from the overall mean. [1]

Overall mean = -0.1646

Mean for Eyes only treatment = -0.26125

Treatment effect = $-0.26125 - (-0.1646) = -0.09665$ [1]

1.2 State the null and alternative hypotheses associated with the F-test in the ANOVA table in as much detail as possible.(1)

Null: There is no difference in the mean melatonin production between the treatments of eyes only, control and knees. [0.5]

Alternative: At least one treatment (eyes only, control or knees) resulted in a different mean melatonin production time. [0.5]

1.3 Using the p-value approach, is there evidence to reject the null hypothesis for the test? Make sure to state your conclusion in the context of the research question. (2)

The p-value is quite large [0.5] therefore we do not reject the null [0.5]. We conclude that there is no difference in the mean melatonin production across the treatments. [1]

1.4 If you repeated this experiment with a much larger sample, what effect would that have on the p-value (assuming the same means and variances)? (1)

Increasing sample size reduces MSE which increases the F-statistic (0.5), and thus decreases the p-value (0.5)

QUESTION 2

[4]

An industrial lab conducted an experiment to compare the scratch resistance of four protective paint coatings applied to metal panels. Since temperature and humidity can vary from one day to the next and influence how the coatings cure, the experiment was conducted over five consecutive days. Each coating (C1–C4) was applied once per day, with the assignment of coatings to panels randomised within each day. After drying, the *scratch resistance* (in Newtons) was measured using a standard abrasion tester. The purpose of the study was to assess whether the average scratch resistance differs among the coatings. See the data and partially complete ANOVA table below.

Day	C1	C2	C3	C4
1	51.4	62.8	54.8	44.6
2	49.4	56.1	54.3	55.6
3	49.2	52	53.6	49.1
4	55.3	62.6	54	47.1
5	47.1	56.7	51	44.1

Source	Df	SS	MS	F
Coat	3	276.04	92.01	7.446

Day	A	B	C	D
Residual	12	148.28	12.36	-
Total	-	495.17		

2.1 Complete the ANOVA by calculating the values of A, B, C and D. (2)

$$A = 19 - 3 - 12 / 5 - 1 = 4 \text{ [0.5]}$$

$$B = 495.17 - 148.28 - 276.04 = 70.85 \text{ [0.5]}$$

$$C = 70.85 / 4 = 17.71 \text{ [0.5] Mark with error}$$

$$D = 17.71 / 12.36 = 1.432.. \text{ [0.5] Mark with error}$$

2.2 Using the results you obtained in 2.1, decide whether it was necessary to conduct the experiment over five consecutive days. Provide a reason for your answer. (1)

No [0.5], blocking (conducting over multiple days) did not improve the experiment as the F test statistics is quite close to one [0.5].

Mark with error if incorrect F-value is calculated, i.e. if $F \gg 1$ then blocking was necessary.

2.3 What assumption is made about the treatment effects in this model? (1)

They are consistent across the blocks / they do not change depending on which block they are in / the treatment effects and block effects are additive [1]

QUESTION 3

[4]

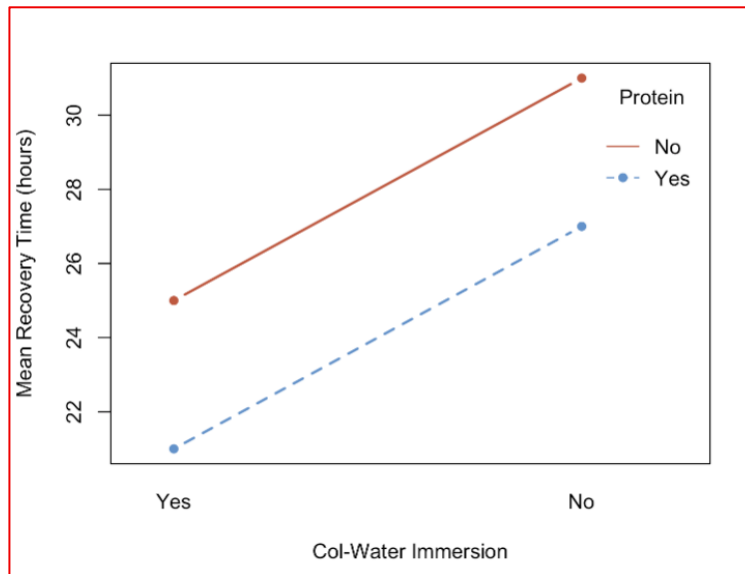
A sports physiologist investigated whether cold-water immersion and protein supplementation influence muscle recovery time after intense training. Athletes were randomly assigned to one of four groups formed by combining the two recovery methods (Cold-water vs No immersion) and Protein intake (Supplement vs No supplement). The outcome variable was time to full recovery (hours) measured using performance tests. In the table below, the mean recovery time is shown.

Cold Water	Protein	Recovery Time
No	No	31
Yes	No	25
No	Yes	27
Yes	Yes	21

3.1 Draw an interaction plot with the available data:

(3)

- Place the factor Cold Water type on the x-axis.
- Draw separate lines for each level of Protein and label each line clearly.



Loose
following instructions (i.e. wrong way around)

one mark for not

Loose a half for something extra like not labelling lines of axes

3.2 Based on the shape of the plot, provide a brief comment on whether an interaction appears to be present. (1)

The lines in the interaction plot are parallel (0.5) which indicates an interaction effect (0.5).

Only marking with error if the interaction plot is sensible.

SECTION C: NON-PARAMETRIC QUESTIONS

[10]

QUESTION 1

[6]

A psychology student wanted to know whether people who drink more coffee in the morning tend to report feeling more alert. Ten participants were asked to record the number of cups of coffee consumed before 10 a.m. and then rate their morning alertness on a scale from 1 (very drowsy) to 10 (very alert). The relationship between coffee intake and alertness was analysed using Spearman's rank correlation coefficient in R and the following output was generated.

Participant	Coffee cups	Alertness
1	4	10
2	3	5
3	3	8.7
4	1	5.4
5	0	3.5
6	5	10
7	2	7.3
8	1	7.5
9	4	7.2
10	0	4.2

1.1 Identify the data type of the two variables measured in this experiment. (1)

The number of coffee cups consumed is a ratio-scaled variable. (0.5)

The alertness score is an interval (mark for ordinal was given as well) variable. (0.5)

1.2 Perform the Spearman's rank correlation test using a 5% significance level. Marks will be awarded for:

1.2.1 Stating the correct null and alternative hypotheses in context of the experiment. (1)

Null: There is no association between the number of coffee cups consumed and alertness. (0.5)

Alternative: There is appositive association between the number of coffee cups consumed and alertness. (0.5)

1.2.2 Calculating the value of the test statistic. (3)

CoffeeCups	Alertness	CupsRank	AlertRank	DiffRank
4	10.0	8.5	9.5	-1
3	5.0	6.5	3	3.5
3	8.7	6.5	8	-1.5
1	5.4	3.5	4	-0.5
0	3.5	1.5	1	0.5
5	10	1	9.5	0.5
2	7.3	5	6	-1
1	7.5	3.5	7	-3.5
4	7.2	8.5	5	3.5
0	4.2	1.5	2	-0.5

$$\sum diff rank_i^2 = 42$$

$$r = 0.7454 \dots$$

$$z = r\sqrt{n-1} = 0.745 \dots \times \sqrt{(9)} = 2.236 \dots$$

1.2.3 Making the correct decision using the p-value approach and a 5% significance level. (1)

p-value = 1 - 0.9875 = 0.0125 [1] < 0.05 [0.5] so we reject the null hypothesis [0.5] .

QUESTION 2

[4]

A food scientist investigated whether three different formulations of a new plant-based burger differ in overall taste. Because people vary widely in how they perceive flavour and texture, each panellist tasted all three burger formulations (F1, F2, and F3) in random order during the same session. Each panellist rated the taste of each sample on a 10-point scale, where higher scores indicate greater overall liking.

Researchers analysed the data using a non-parametric test obtaining a test statistic of 5.02 and a p-value of 0.08.

5.1 Explain why a non-parametric test might be preferred in this experiment instead of an ANOVA. (1)

Because taste ratings are on an ordinal scale (1)

5.2 Name the correct non-parametric test that should be used for this design, and state its null and alternative hypotheses in context. (2)

Friedman test [1]

Null: There is no difference in the taste ratings between the plant-based burgers. [0.5]

Alternative: There is a difference in the ratings between the plant-based burgers. [0.5]

5.3 Why is it inappropriate to perform three separate Wilcoxon signed-rank tests (F1 vs F2, F2 vs F3, F1 vs F3) instead of using a single test? (1)

Because doing multiple pairwise tests inflates the Type I error rate. (1)

SECTION D: TIME-SERIES

[11]

QUESTION 1

1.1 For ANY TWO of the four components that we find in a time series, provide a concise one-sentence but *detailed* definition. Your description for each component should start on a new line. (2)

The trend (0.5) is the overall/long-term movement of the data (0.5)

Seasonality (0.5) is regular, short-term, repetitive, wave-like variation. (0.5) (must have short-term & repetitive to get 0.5)

Cyclical (0.5) is irregular, long-term (deviation from trend) wave-like variation. (0.5) (must have irregular and long-term to get 0.5)

Random variation (0.5) refers to variation due to unforeseen/unpredictable events. (0.5)

1.2 Define a stationary time series. (1)

A time series that has constant mean (0.5) and constant variance/variation (0.5)

OR:

a time series where the mean and variance (0.5) are not changing with time

a time series where the mean and variance (0.5) are independent of time. (0.5)

OR: statistical properties of the time series remain constant/independent of time

0.5 for saying no trend or seasonality

- 1.3 Suppose you have an hourly time series that displays decreasing linear trend, seasonality (where the length of a seasonal variation is one day), constant variance and random variation. How would you transform this time series to make it stationary? (1)
Difference the data (0.5) at lag 24 (0.5)

Questions 1.4 to 1.7 below relate to the following context:

Suppose you have time series data on the daily total logins to the STA2020S 2025 Amathuba site for the first 6 weeks of the semester. You are certain that there will be seasonality present in this time series. You decide to conduct a time series analysis.

- 1.4 What would be your *first* and *second* step in your time series analysis? (2)

Plot the time series data. (1)

Try to identify what components are present in the time series. (1)

(0.5) assessing stationarity

(0.5) what is the length/how many time periods in a seasonal variation.

You conduct a classical decomposition on the time series.

- 1.5 What value of k would you choose for the $MA(k)$ smoothing, and why would this be appropriate?. (1)

$k = 7$ (0.5), since the seasonality is likely to last one week/7 days OR repeat each week/7 days/7 days in a week) (0.5)

- 1.6 Suppose your adjusted seasonal index for Sundays is 0.67. How would one interpret this value? Be specific and detailed in your interpretation. (2)

In general/on average (0.5), the daily total visitors to the Amathuba site on Sundays/on that day (0.5) are 33% below (0.5) the weekly/seasonal average (0.5).

You decide to make some short-term forecasts using a simple forecasting method. Your data is saved as the object `site_visits`.

- 1.7 Which simple forecasting method would you use to make predictions for the next two weeks, and what would be the R Code to apply that method (with the actual names/values for the arguments). (2)

Seasonal naïve (1) method

`snaive(site_visits,14)` OR `snaive(data = site_visits,h = 14)` (1)

(0.5) for `snaive`

Mark with the error if they gave the wrong method.

STA2020F: Important Formulae

SIMPLE LINEAR REGRESSION

Sum of squares:

$$SS_{reg} = SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

$$SS_{resid} = SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$SS_{tot} = SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

Measures of association:

$$r_{xy} = \frac{SS_{xy}}{\sqrt{SS_x SS_y}}$$

Standard error of the regression slope:

$$s_{\beta_1} = \frac{s_{\varepsilon}}{\sqrt{SS_x}}, \text{ where } RSE = s_{\varepsilon}$$

$(1 - \alpha)\%$ confidence interval for β_j :

$$\hat{\beta}_j \pm t_{\frac{\alpha}{2}, n-p-1} s_{\beta_j}$$

MULTIPLE REGRESSION

Multiple regression model (for the sake of defining n and p):

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i,1} + \hat{\beta}_2 x_{i,2} + \cdots + \hat{\beta}_p x_{i,p} \text{ for } i = 1, 2, 3, \dots, n$$

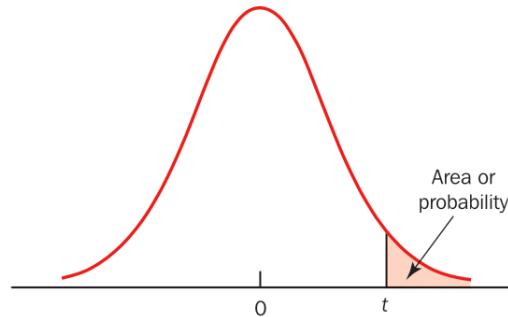
LOGISTIC REGRESSION

logit: $\log\left(\frac{p}{1-p}\right)$

Inverse logit: $p = \frac{1}{1+\exp^{-(\beta_0+\beta_1x+\cdots)}}$

$(1 - \alpha)\%$ confidence interval for β_j :

$$\hat{\beta}_j \pm Z_{\frac{\alpha}{2}} s_{\beta_j}$$

TABLE 2 *t* distribution

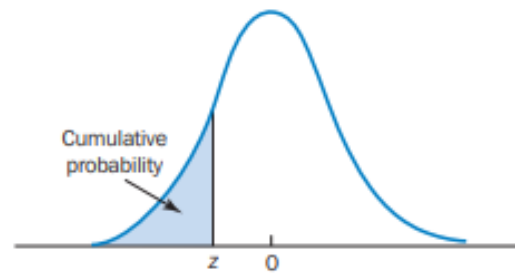
Entries in the table give *t* values for an area or probability in the upper tail of the *t* distribution. For example, with ten degrees of freedom and a .05 area in the upper tail, $t_{.05} = 1.812$.

Degrees of freedom	Area in upper tail					
	.20	.10	.05	.025	.01	.005
1	1.376	3.078	6.314	12.706	31.821	63.656
2	1.061	1.886	2.920	4.303	6.965	9.925
3	.978	1.638	2.353	3.182	4.541	5.841
4	.941	1.533	2.132	2.776	3.747	4.604
5	.920	1.476	2.015	2.571	3.365	4.032
6	.906	1.440	1.943	2.447	3.143	3.707
7	.896	1.415	1.895	2.365	2.998	3.499
8	.889	1.397	1.860	2.306	2.896	3.355
9	.883	1.383	1.833	2.262	2.821	3.250
10	.879	1.372	1.812	2.228	2.764	3.169
11	.876	1.363	1.796	2.201	2.718	3.106
12	.873	1.356	1.782	2.179	2.681	3.055
13	.870	1.350	1.771	2.160	2.650	3.012
14	.868	1.345	1.761	2.145	2.624	2.977
15	.866	1.341	1.753	2.131	2.602	2.947
16	.865	1.337	1.746	2.120	2.583	2.921
17	.863	1.333	1.740	2.110	2.567	2.898
18	.862	1.330	1.734	2.101	2.552	2.878
19	.861	1.328	1.729	2.093	2.539	2.861
20	.860	1.325	1.725	2.086	2.528	2.845
21	.859	1.323	1.721	2.080	2.518	2.831
22	.858	1.321	1.717	2.074	2.508	2.819
23	.858	1.319	1.714	2.069	2.500	2.807
24	.857	1.318	1.711	2.064	2.492	2.797
25	.856	1.316	1.708	2.060	2.485	2.787
26	.856	1.315	1.706	2.056	2.479	2.779
27	.855	1.314	1.703	2.052	2.473	2.771
28	.855	1.313	1.701	2.048	2.467	2.763
29	.854	1.311	1.699	2.045	2.462	2.756

TABLE 2 (Continued)

Degrees of freedom	Area in upper tail					
	.20	.10	.05	.025	.01	.005
30	.854	1.310	1.697	2.042	2.457	2.750
31	.853	1.309	1.696	2.040	2.453	2.744
32	.853	1.309	1.694	2.037	2.449	2.738
33	.853	1.308	1.692	2.035	2.445	2.733
34	.852	1.307	1.691	2.032	2.441	2.728
35	.852	1.306	1.690	2.030	2.438	2.724
36	.852	1.306	1.688	2.028	2.434	2.719
37	.851	1.305	1.687	2.026	2.431	2.715
38	.851	1.304	1.686	2.024	2.429	2.712
39	.851	1.304	1.685	2.023	2.426	2.708
40	.851	1.303	1.684	2.021	2.423	2.704
41	.850	1.303	1.683	2.020	2.421	2.701
42	.850	1.302	1.682	2.018	2.418	2.698
43	.850	1.302	1.681	2.017	2.416	2.695
44	.850	1.301	1.680	2.015	2.414	2.692
45	.850	1.301	1.679	2.014	2.412	2.690
46	.850	1.300	1.679	2.013	2.410	2.687
47	.849	1.300	1.678	2.012	2.408	2.685
48	.849	1.299	1.677	2.011	2.407	2.682
49	.849	1.299	1.677	2.010	2.405	2.680
50	.849	1.299	1.676	2.009	2.403	2.678
51	.849	1.298	1.675	2.008	2.402	2.676
52	.849	1.298	1.675	2.007	2.400	2.674
53	.848	1.298	1.674	2.006	2.399	2.672
54	.848	1.297	1.674	2.005	2.397	2.670
55	.848	1.297	1.673	2.004	2.396	2.668
56	.848	1.297	1.673	2.003	2.395	2.667
57	.848	1.297	1.672	2.002	2.394	2.665
58	.848	1.296	1.672	2.002	2.392	2.663
59	.848	1.296	1.671	2.001	2.391	2.662
60	.848	1.296	1.671	2.000	2.390	2.660
61	.848	1.296	1.670	2.000	2.389	2.659
62	.847	1.295	1.670	1.999	2.388	2.657
63	.847	1.295	1.669	1.998	2.387	2.656
64	.847	1.295	1.669	1.998	2.386	2.655
65	.847	1.295	1.669	1.997	2.385	2.654
66	.847	1.295	1.668	1.997	2.384	2.652
67	.847	1.294	1.668	1.996	2.383	2.651
68	.847	1.294	1.668	1.995	2.382	2.650
69	.847	1.294	1.667	1.995	2.382	2.649

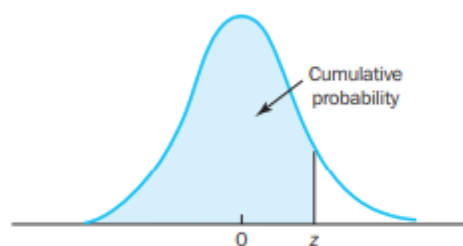
(continued)

TABLE 1 Cumulative Probabilities for the Standard Normal Distribution

Entries in the table give the area under the curve to the left of the z value. For example, for $z = -.85$, the cumulative probability is .1977.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

TABLE 1 (Continued)



Entries in the table give the area under the curve to the left of the z value. For example, for $z = 1.25$, the cumulative probability is .8944.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9913
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9986	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990